

Course III: Corporate Finance

Summer Term 2026

Unit 1 Mini-Cases

Please upload your solutions and indicate the Mini-Cases you have solved before 10:00 p.m. on March 15, 2026. There will be no deadline extensions!

To upload your solutions and make indications, please go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). Only complete submissions containing answers to the T/F questions and a solutions upload will be considered. Your last complete attempt counts.

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solution file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Mini Case 1.1: Book values and market values

Choose one publicly traded company from each of the Technology and Utilities sectors included in the S&P 500. Consult the most recent annual reports of the firms. You may need to consult additional reliable data sources. Please include a link to the data sources you use (including to the financial report).

- a. Find the total assets, liabilities, and book value of equity and debt for both companies.
- b. Find the number of outstanding shares for the company and its share price as of December 31, 2025, for both companies.
- c. Calculate the market value of the company's equity, i.e. the market capitalization for both companies.
- d. Estimate the market value of total assets for both companies. (Hint: Use the book value of debt as a proxy for the market value of debt.)
- e. Calculate the Market-to-Book Ratio for both companies and
 - i. explain where potential differences between market and book values stem from.
 - ii. explain the structural differences in the ratios between the sectors.

Mini Case 1.2: Lease vs. Buy

Lemon Inc. (LI) currently spends \$120,000 per year leasing office space. The lease Payments are tax-deductible, and the corporate tax rate is 30%. The firm has no debt and an equity beta of 1.4. Assuming an expected market return of 11% and a risk-free rate of 3%, its CAPM-based cost of equity capital is 16.5% (not directly used for the building decision).

LI has the opportunity to buy the office space for \$2.2 million. The building is relatively risk-free, and the company can finance 100% of the purchase with tax-deductible mortgage payments. The mortgage rate is 4.0%. Assume the building is fairly priced at \$2.2 million and that LI plans to use it in perpetuity.

- a. Should Lemon Inc. buy the building or continue to lease it?
- b. What is LI's maximum willingness to pay for the building?
- c. If LI's corporate tax rate were 0% (no tax shield), would it still be better to buy rather than lease?

Mini Case 1.3: Leverage

A firm's assets have a market value of EUR 224 Mio. The firm is financed with debt and equity.

- a. Suppose the market value of debt is EUR 160 Mio.
 - i. What is the market value of equity?
 - ii. Suppose the market value of assets decreases by 10%. What is the market value of equity? By what percentage has it changed?
 - iii. Suppose the market value of assets increases by 20%. What is the market value of equity? By what percentage has it changed?
- b. Instead, suppose the market value of debt is only EUR 40 Mio.
 - i. What is the market value of equity?
 - ii. Suppose the market value of assets decreases by 10%. What is the market value of equity? By what percentage has it changed?
 - iii. Suppose the market value of assets increases by 20%. What is the market value of equity? By what percentage has it changed?
- c. Based on your calculations, how does debt affect the riskiness of equity?

Mini Case 1.4: Equity and Debt Beta

A company produces a perpetual stream of expected free cash flows of 100 per year from next period onwards. The realized cash flows are risky and have an asset beta of 0.8. The company is levered with a debt-equity ratio of 1. The risk-free rate is 4% and the market return is 10%.

Assume Modigliani-Miller holds.

- a. Calculate the asset return (Note that $r_A = r_U$)
- b. Calculate the firm value and the WACC
- c. Assume that debt is risk free and calculate
 - i. Equity beta
 - ii. Cost of equity
 - iii. Cost of Debt
 - iv. WACC
- d. Assume that debt is risky with a debt beta of 0.4 and calculate
 - i. Equity Beta
 - ii. Cost of Equity
 - iii. Cost of Debt
 - iv. WACC
- e. Explain the changes for debt and equity holders between c) and d). What is the impact on firm value?

Mini Case 1.5: Arbitrage and MM

Two firms, U and L, produce identical cash flows next period of either 320 or 140 (equal probability) and no cash flow thereafter.

- U is 100% equity financed with firm value $V(U) = 250$.
 - L is financed with debt and equity. L's debt is priced at 80 and L's equity at 190, so $V(L) = 270$.
- a. Is there an arbitrage opportunity (assume perfect capital markets except for the prices given)?
 - b. If so, how would you exploit it? State clearly all transactions and cash flows in $t=0$ and $t=1$. The risk-free rate is 5%.
 - c. How do you expect prices to change?

Mini Case 1.6: Who benefits from a positive NPV project?

Ben's Coffee Company (BCC) has won a licence to offer artisan coffee from a van outside the Schloss Schönbrunn. BCC estimates that the present value of the costs of operating in the first year are EUR 500 (to hire the van, purchase ingredients, and pay wages for staff etc.). If the weather is good, there will be lots of tourists and BCC's total revenue will be EUR 900 (future value). If the weather is bad, there will be fewer tourists, and revenue will be EUR 400 (future value). The chance for good or bad weather is 60% and 40% respectively. BCC has to decide whether to offer coffee or not before the weather becomes known. Tourism does not depend on the performance of the stock market or the aggregate economy, but it does depend on the weather. All parties are risk-neutral, and the risk-free rate is 4% p.a.

- a. What is the NPV of the licence to offer coffee?
- b. Suppose that BCC has no funds and therefore plans to raise financing by borrowing from capital markets, i.e., by issuing debt that is to be repaid after one year. There is competition in capital markets, so BCC can obtain debt financing at the fair price. It is conventional in this market to not demand interest payments in the interim periods. The bond commits BCC to repay the face value of the bond conditional on BCC not entering bankruptcy, and to hand over all assets if it enters bankruptcy. If BCC wants to raise EUR 500, what face value does it need to specify on the bond?
- c. What is the value of the BCC's equity after taking on debt finance to fund the project?